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Using artificial intelligence to predict patient wait times in the emergency department: A scoping review

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ABSTRACT

Objective: The purpose of this review was to comprehensively explore the landscape of recently published literature on the applications of artificial intelligence (AI) in predicting individualized patient waiting times in an emergency department (ED) and identify pertinent considerations for practitioners and hospital decision-makers.

Introduction: ED overcrowding is being experienced by hospitals around the globe and has worsened in the post COVID-19 era. The negative patient and staff experiences and poor clinical outcomes from overcrowding are evident and necessitate solutions to address this ongoing problem. Hospitals providing ED waiting time estimates to patients and staff are becoming popular; however, the more common methods, such as using rolling averages, suffer from an inability to capture the nuanced relationships within an ED. Recent applications of AI and machine learning (ML) in healthcare raises the possibility of applying these techniques to individualized waiting time predictions in the ED; although, literature on the topic is sparse.

Methods: A systematized search was conducted on November 10th, 2025, using the electronic databases CINAHL, EMBASE (OVID), Medline (OVID), PsychINFO, Web of Science, and PubMed. Articles were considered for review if written in English, peer-reviewed, published after 2014, and used AI techniques. Descriptive analysis was performed on the final extracted data to facilitate the identification of common themes across studies. Themes were inferred from the proportional usage among studies, of different data preparation, feature selection, and modeling strategies.

Results: The search identified 8613 citations that, after a rigorous screening process and critical appraisal, were narrowed down to 15 studies for final review. Most included studies were observational, using historical medical record data to compare modeling techniques or demonstrate a proof of concept. Studies commonly used one or more of ED queue-based, staff/resource-based, patient-based, and time-based feature categories. Incorporated AI methods included Random Forest, Linear Regression, and Least Absolute Shrinkage and Selection Operator (LASSO) techniques, among several others. All forms of AI and ML outperformed traditional rolling average estimates used by hospitals.

Conclusions: This review identified applications of AI in predicting individualized patient waiting times in the ED that outperform current waiting time estimate strategies. The use of nonlinear techniques, such as the Random Forest method, or incorporating queue-based feature categories, appeared to provide better performance in predictive estimates. Depending on the end user and modality in which the wait time estimate is conveyed, the importance of model selection is highlighted as a consideration to be made if overestimates or underestimates are preferred.

Abbreviation: AI, artificial intelligence; ED, emergency department; EHI, electronic health information; EHR, electronic health record; Long-Short Term Memory, LSTM; ML, machine learning.

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1. Introduction

Overcrowding in an emergency department (ED) is a common global concern, as it negatively affects patient outcomes and experiences [1], staff experience, and strains healthcare resources [2]. ED overcrowding occurs when the demand for healthcare services exceeds the capacity of an ED to provide quality care within appropriate time frames [3]. Globally, some hospitals have experienced increased ED overcrowding following the COVID-19 pandemic [4,5]. The urgency to address this ongoing problem is emphasized by overcrowding leading to patients with more dangerous and time-sensitive conditions to experience prolonged delays in receiving care, depart the ED before completing treatment, or avoiding the ED altogether [6].

As a result of EDs increasingly incorporating information technology into workflow processes, electronic systems have enabled the routine capture of timestamps along the patient journey. Standardized interval metrics have emerged from these timestamp records as reference points for evaluating operational and quality improvement strategies [7]. Patient waiting time, typically defined from registration or triage to physician initial consult, is one such important metric that has implications for ED overcrowding levels, patient condition, and incremental costs of care to the hospital [8]. Waiting time has been used by ED decision makers to optimize staffing decisions [9]. Moreover, the knowledge of patient waiting times empowers staff to make informed care pathway decisions for patients in the ED, potentially improving departmental flow. Alternatively, from a patient perspective, it has been demonstrated that satisfaction of healthcare services is improved when patients receive information on their expected waiting time [10]. Also, presenting waiting time estimates to patients continually throughout their stay might improve waiting tolerance, reducing the incidence of patients leaving without being seen by a physician [11]. Some hospitals have therefore opted to display estimated waiting times in the ED. Extending beyond the waiting room, having real-time waiting time estimates available to patients within a network of hospitals empowers them to make informed routing decisions and alleviates local ED overcrowding via patient load balancing [12]. These use cases create incentives for identifying methods to predict patient waiting times for both hospital and patient use.

Common strategies that hospitals have used to predict wait times include simpler statistical techniques such as rolling average methods and autoregressive integrated moving averages. It is noteworthy however, that the simplistic nature of these methods leads them to be inaccurate and misleading in some cases [13]. Due to the accumulation of detailed electronic health information (EHI) and increasingly more capable hardware systems, there has been growing interest in applying more sophisticated artificial intelligence (AI) techniques to the task of predicting ED wait times. Accommodating multivariate linear or non-linear dependencies appears to enhance predictive potential in complex real-world problems such as within an ED [14]. The expanding repository of EHI contains many different patient-related, ED queue state, and temporal feature data that can be leveraged to create a predictive model that captures more of the nuance involved in the stochastic ED environment. This also allows for more individualized wait time predictions.

Because the term AI encapsulates a broad spectrum of concepts that involve creating computer and robotic systems to perform tasks typically requiring human intelligence, this paper specifically refers to AI in the context of machine learning (ML). Predictive modeling techniques within this definition can be described as 'linear' or 'non-linear' depending on the assumptions of the relationship between the input variables and the target. Linear modeling techniques are sometimes preferred because of their ease of understanding, interpretation, and low computational requirements [15]. Non-linear strategies are beneficial when the underlying data structure is complex, however suffer from interpretability due to the complexity, and are commonly referred to as 'black box' models [16]. Either technique may be preferred depending on the context of the problem, and both are generally evaluated prior to implementing a solution.

This scoping review was conducted to probe the breadth of research on the current applications of using AI to predict individualized patient wait times in an ED setting. The strategy involved identifying and evaluating published peer-reviewed studies where individualized patient waiting times were forecasted using AI techniques. Specifically, we aim to review the current applications of AI for predicting individualized patient wait times in hospital EDs.

The justification for conducting this scoping review is underscored by the fact that no reviews on the topic have been performed to date. There is literature that exists which encompasses a broader scope including all applications of AI in an ED [17], in addition to adjacent reviews that focus on AI and other statistical methods for ED length of stay metrics rather than wait times [18] or using AI in triage [19]. This research is therefore timely and worthwhile, as it contextualizes the strategies utilized thus far for patient wait time prediction using AI. Moreover, it provides a succinct resource for hospital management and other researchers to glean an understanding on the scope of research on the topic which can support development and implementation of site-specific wait time prediction models, or to stimulate future research directions. Lastly, this research captures the contribution of advances in AI to ED wait time prediction that ultimately can improve ED workflow management, patient satisfaction and outcomes, and alleviate the burden of overcrowding within the ED.

2. Material and methods

This scoping review was conducted following the Joanna Briggs Institute methodology for scoping reviews as depicted in the PRISMA diagram in Fig. 1 [20].

2.1. Eligibility criteria

Population: This review imposed no geographical constraints, which yielded an approach ensuring a comprehensive analysis without exclusive focus on specific population subsets.

Concept: The focus of this review were studies that used ML (e.g., regressions, decision trees, artificial neural networks) to predict or understand wait times in the ED. Studies that focused on designing an ED process or policy for improving patient wait times were excluded. Additionally, studies that predicted patient waiting times aside from arrival to physician consult, admission, or overcrowding were excluded. Studies that focused on improving triage/screening processes (e.g., risk analysis, measures) and not wait times directly were also excluded.

Context: Studies that were not published in English were excluded. To maintain relevancy to current literature, only studies published since 2014 were considered.

Types of studies: The review included a variety of studies, including peer-reviewed randomized controlled trials, observational studies, pilot studies, and case reports. Systematic reviews, literature reviews, meta-analyses, opinion pieces, commentaries, dissertations, and letters to the editor were excluded from this review.

2.2. Search strategy

Our research team utilized six databases (CINAHL, EMBASE (OVID), Medline (OVID), PsychINFO, Web of Science, and PubMed) to retrieve articles on predicting wait times in the ED. The finalized search strategy, which was used for the OVID MEDLINE database, and the inclusion/exclusion criteria can be found in the attached Supplementary Materials.

2.3. Study selection

The results produced from the finalized search strategy were imported into the Covidence systematic review software (Veritas Health

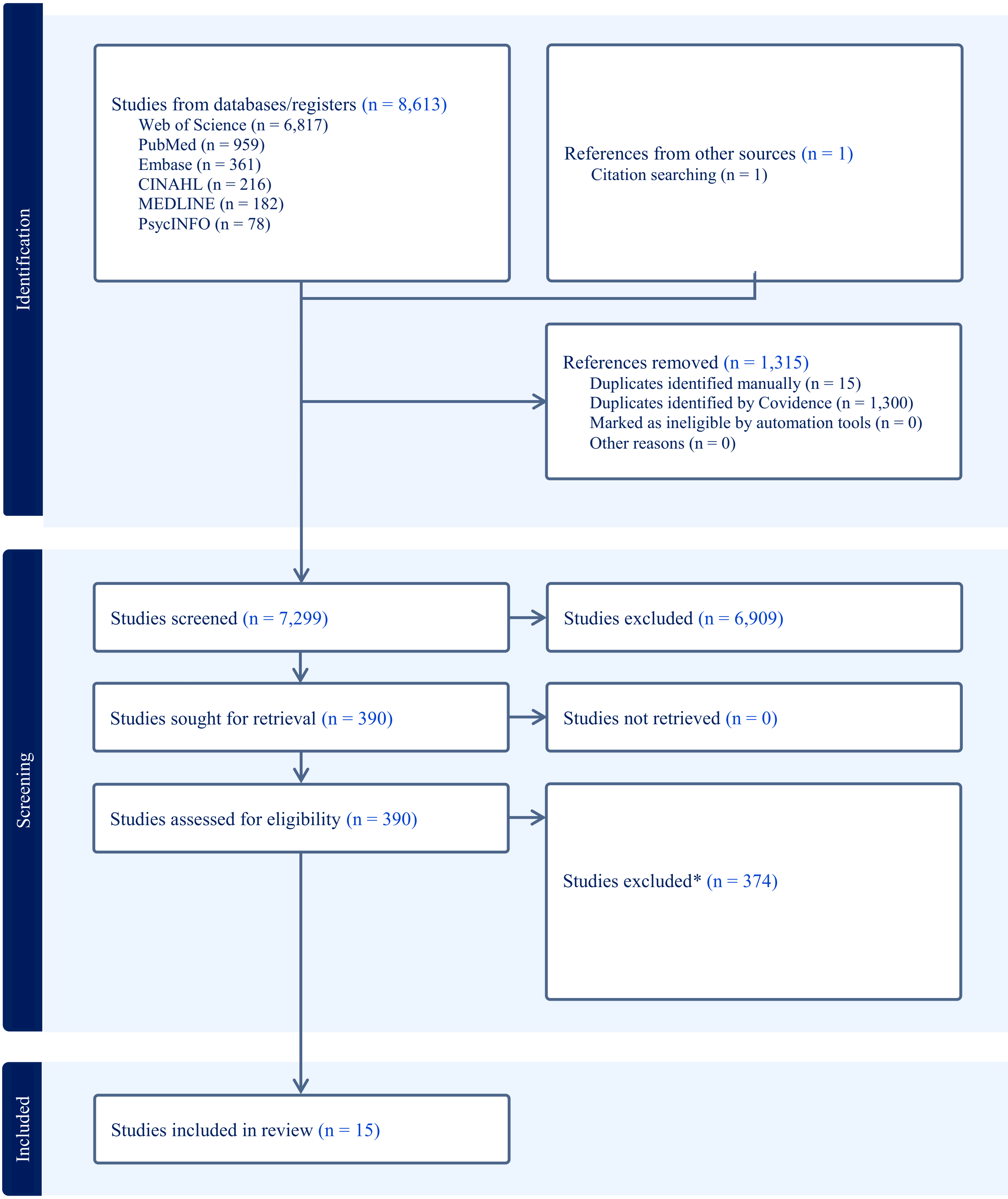

Fig. 1. PRISMA flow diagram showing identification, screening, and included studies. *Detailed breakdown of exclusion criteria in the Supplementary Materials.

Innovation, Melbourne, Australia). All duplicate studies were identified and withdrawn. Three authors completed the initial title and abstract screening stage, consisting of 7299 articles. Studies that were included were then moved to the full-text review stage (390). Studies not located in the reviewers' university library were requested directly from study authors. In the full-text review stage, three reviewers independently screened each article using the inclusion/exclusion criteria. Any disagreements that were raised during this stage were later resolved through a group discussion between the three reviewers and two additional reviewers. Under circumstances of reviewer unavailability, only two independent reviewers screened and resolved disagreements via consensus discussions. One article that was found post-hoc via examining the reference lists of selected studies is included in all subsequent analyses.

2.4. Data extraction

The reviewers developed the extraction template together to ensure it included all the variables our study aimed to evaluate. One author and co-reviewers tested out the extraction template to ensure that the data extraction information was feasible. Data extraction for each study was completed by two independent reviewers. Any conflicts that emerged during the extraction stage were resolved by a third reviewer to reach a consensus. Under circumstances of third reviewer unavailability, conflicts were discussed and settled between two reviewers in consensus meetings. The information extracted from each article can be found in the attached Supplementary Materials.

2.5. Data analysis

The results from the data extraction were exported into Microsoft Excel and converted into several data tables for ease of comparison between studies. Descriptive statistics were performed to evaluate counts and proportions. From these statistics, broad themes were identified pertaining to the dataset preprocessing, feature inclusion, and modeling strategies.

3. Results

3.1. Study summary statistics

Table 1 displays summary statistics from the included articles. Considering Wang [35] is an extension of Wang [34] that includes a model bias assessment, and is methodologically similar with regards to the dataset, model development and feature inclusion, they are herein grouped together for study accounting purposes. Most (14/15) of the studies were retrospective, including patient visit records to an ED. The exception was one study that created synthetic patient timestamp data which was subsequently fed into a Linear Regression and a Long-Short Term Memory (LSTM) model [23]. Studies were conducted in ten different countries and spanned the publication years 2016 to 2025. The fact that 12/15 studies were published in the last 5 years suggests that there is growing interest in this topic. Where historical electronic health record (EHR) datasets were used, the sample size to train and test the models ranged from 12,023 [24] to 1,930,609 [27]. These datasets covered timepoints from the years 2011 [21] to 2022 [28] and spanned timeframes from as short as 1 month [26] to 60 months (about 5 years) [30]. Data cleaning strategies focused mainly on removing patient records with null values, incorrect entries, or outliers. Out of the 15 studies that used EHR data, eight provided detailed information on the extent of data removal, as a percentage dropped, or number of records removed. Five studies incorporated data from multiple hospitals [21,27,29,30,32], while others performed analysis on data from a single site. Decisions to include multiple hospital sites included model performance replicability verification with different data populations [21,27,29,30,32], external model validation [27], and a conceptual demonstration of remotely forecasting wait times to patients from within a network of hospitals [30]. Five studies decided to exclude patients with a higher acuity and predicted wait times for only low acuity patients [21,24,25,30,31,34,35]. Patient acuity was generally assessed from a categorical severity scale measured following nurse triage.

3.2. Incorporated wait-time predictors

There were common themes that arose between the features incorporated in each study that, for simplicity, were grouped into five distinct categories and included: queue-based features, staff/resource-based features, patient-based features, time-based features, and an ‘other’ features category. Feature inclusions in each study are summarized in Table 2. The ‘other’ category contained features that weren’t easily grouped into the previous four categories. Queue-based features were commonly the number of patients waiting at different points in their care pathway (e.g., waiting for triage, consultation, discharge by acuity level). Staff/resource-based features included the number of physicians and nurses on shift, beds and room capacities, and specific care department availabilities. Patient-based features typically included age,

Table 1
Descriptive highlights of the 15 included studies in the scoping review.

Study	Country	Year of study	Dataset timeframe (Mo)	# Records used	Multi-site	Patient subset	Performance measures	End user
Ang 2016 [21]	USA	2011–2015	12–16	57,823 - 122,326	Yes	Low Acuity	MSE	Patient
Benevento 2019 [13]	Italy	2017	7	34,676	No	NR	MSE	Both
Goncalves 2019 [22]	Portugal	2013–2017	48	672,720	No	No	Precision, True Positive Rate, F1-Score	Hospital
Cheng 2020 [23]	Canada	NR	NA	50,000	NA	NA	MAE	Hospital
Kuo 2020 [24]	China	NR	1	12,023	No	Low Acuity	R-Squared, MSE	Both
Pak 2021 [25]	Australia	2016–2017	24	94,284	No	Low Acuity	MAPE, MSPE	Patient
Ataman 2021 [26]	Turkey	2018	1	37,711	No	NR	Accuracy	Both
Walker 2022 [27]	Australia	2017–2020	36	1,930,609	Yes	No	AE, Median Absolute Error	Patient
Ameur 2022 [28]	France	2019–2022	39	88,166	No	No	MAE, RMSE	Hospital
Benevento 2023 [29]	Italy	NR	8–12	38,081 - 57,887	Yes	No	MAE, MSE	Patient
Arora 2023 [30]	UK	2014–2018	60	352,178	Yes	Low Acuity	CRPS, RPS, RMSE, MAE	Patient
Guo 2023 [31]	China	2021	12	183,024	No	Low Acuity	MAE, RMSE	Hospital
Cancellas 2024 [32]	Germany	2019–2020	12	123,381	Yes	No	Accuracy, F1-Score, AUC, Sensitivity, Specificity	Hospital
Hogben 2025 [33]	UK	2022	12	40,828	No	No	Success Rate	Patient
Wang 2025 [34,35]	USA	2019–2021	36	177,665, 173,856	No	ESI 3	Accuracy, Recall, Precision, F1-Score, AUROC, Counterfactuals	Hospital

Notes: NR = Not reported; NA = Not applicable; MSE = Mean squared error; MAE = Mean absolute error; RMSE = Root mean squared error; MAPE = Mean absolute percentage error; MSPE = Mean squared percentage error; AE = Absolute error; CRPS = Continuous ranked probability score; RPS = Ranked probability score; AUC = Area under the curve; ESI = Emergency severity index; AUROC = Area under the receiver operating characteristic.

Table 2
Model types and feature categories included in each reviewed study.

Study	RA	FM	LR	LogR	QR	Lasso	Q-Lasso	RR	EN	OLR	SV	KNN	RF	QRF	GBM	ANN	LSTM	EM
Ang 2016 [21]	X	X			X		X*											
Benevento 2019 [13]						X							X*					
Goncalves 2019 [22]													X					
Cheng 2020 [23]			X														X*	
Kuo 2020 [24]			X								X				X*	X		
Pak 2021 [25]			X		X	X*		X					X					
Ataman 2021 [26]										X								
Walker 2022 [27]	X		X*						X				X*					
Ameur 2022 [28]			X			X					X		X*			X*		
Benevento 2023 [29]						X					X		X			X		X*
Arora 2023 [30]	X				X		X					X		X*				
Guo 2023 [31]	X		X			X						X	X		X*			
Cancellas 2024 [32]				X									X		X			
Hogben 2025 [33]			X															
Wang 2025 [34,35]				X							X		X		X*	X		

Notes: RA = Rolling Average; FM = Fluid Model; LR = Linear Regression; LogR = Logistic Regression; QR = Quantile Regression; RR = Ridge Regression; EN = Elastic Net; OLR = Ordinal Logistic Regression; SV = Support Vectors; KNN = K-Nearest Neighbours; DT = Decision Tree; RF = Random Forest; QRF = Quantile Regression Forest; GBM = Gradient Boosting Model; ANN = Artificial Neural Network; LSTM = Long-Short Term Memory Network; EM = Ensemble Method. *Best performing model if comparing more than one.

Study	Feature set categories				
	Queue	Staff	Patient	Time	Other
Ang 2016 [21]	X	X		X	X
Benevento 2019 [13]	X	X	X	X	
Goncalves 2019 [22]		X	X	X	
Cheng 2020 [23]				X	
Kuo 2020 [24]	X	X	X	X	
Pak 2021 [25]	X	X	X	X	X
Ataman 2021 [26]			X		
Walker 2022 [27]	X		X	X	
Ameur 2022 [28]	X		X	X	
Benevento 2023 [29]	X		X	X	
Arora 2023 [30]	X	X	X	X	
Guo 2023 [31]	X	X	X	X	
Cancellas 2024 [32]	X	X	X	X	
Hogben 2025 [33]	X		X	X	
Wang 2025 [34,35]	X		X	X	

sex, acuity level, arrival mode, and occasionally the reason for visit. Time-based features included patient arrival time of day, day of the week, month, triage time, and the average wait time of previous patients. Lastly, features within the 'other' category consisted mostly of external weather information. Of the 15 studies, time-based features were most common (14 studies), closely followed by patient-based features (13 studies) and queue-based features (12 studies). Most studies used a combination of two or more feature categories, except for two studies that used only one [23,26]. Only one study [25] used features from all five categories.

The justifications for feature inclusion varied between studies and included domain expertise, referencing of previous literature, statistical comparative analyses, a stepwise feature hold-out approach, feature importance plots for decision tree-based methods, or no justification at all. For example, Ataman 2021 [26] evaluated the significance of mean wait time difference between the categories of input variables using independent sample *t*-tests and analysis of variance tests (ANOVA). Moreover, [13,21,24,28] used a stepwise feature hold-out method whereby different feature categories were iteratively added to the model to evaluate the changes in performance. Of note, eight studies [13,21,25,27–31] reported using the Lasso, Q-Lasso, and Elastic Net methods which apply a regularization penalty term proportional to the absolute value of the coefficients. Although not explicitly stated in the studies, this acts as an automatic feature selection strategy, driving less important feature coefficients to zero.

3.3. Data splitting, evaluation metrics, and ML modeling

Most studies split their data into a training set and testing set, except three that used all the data for the training set [26,31,35] and one that was unclear [22]. Of those that had a train/test split (12 studies), the training set ranged between 70 % and 80 % of the data. All twelve studies reported their test set results, while only two [24,34] reported their training results. Conducting cross-validation or utilizing a separate validation set was reported in 9/15 studies.

Across studies, model performance was evaluated using 19 different metrics, with Mean Absolute Error (MAE) (5 studies), Mean Squared Error (MSE) (4 studies) and Root Mean Squared Error (RMSE) (3 studies) being the most common. Three studies [13,30,32] validated their methodology by replicating their training framework on a new set of data, however only one study [27] conducted an external validation of their model.

Depicted in Table 2, all studies except for two [22,26] compared multiple different ML modeling techniques to predict wait times, with five being the median number of models used. Five studies [22,26] evaluated wait time prediction as a classification problem, categorizing wait time intervals into distinct groups, and the rest addressed it as a regression problem. The two most common ML techniques were Random Forest (9 studies) and Linear Regression (7 studies). For simplicity, modeling techniques that introduced one or more regularization parameters in a Linear Regression (Ridge, Lasso, Elastic Net) were considered variant models and not counted towards the total. Additionally, a change in the loss function from Ordinary Least Squares to the Huber Loss was not considered a variant model of Linear Regression. Most studies (12/15) used a combination of interpretable models and models that are typically considered as a "black box". Eight out of those 12 studies demonstrated that "black box" models performed best, two [27,32] had comparable results between both, and one [25] showed the interpretable model outperforming. Three studies [24,31,34] used Gradient Boosting Models (considered "black box"), and in each, this technique outperformed the others being evaluated. The Gradient Boosting methods contained several variants including Gradient Boosting Machines [24], LightGBM and XGBoost [31,34]. Importantly, in studies that compared ML methods with the traditional simple statistical technique of a rolling average, the ML methods outperformed every time. A brief description of each modeling method can be found in Table 3 in the Supplementary Materials.

3.4. Wait-times to improve patient experience most common

The utility of the wait time prediction was mentioned as an opportunity to improve patient experience by presenting estimates in the waiting room or online, as a tool for hospital resource allocation decision making, or for ED patient load balancing in a hospital network. When comparing the purpose of the wait time predictions, there were six studies predicting wait times to improve patient satisfaction, six studies for hospital decision making, and three studies that mentioned both applications. Arora 2023 [30] was the only study to demonstrate the concept of patient load balancing in a hospital network by presenting online wait time estimates. Three studies [21,25,30] discussed implementation plans, and all were applications for improving patient experience.

4. Discussion

This scoping review identified 15 studies incorporating applications of AI to predict individualized ED patient waiting times using EHR data mining approaches and artificially generated patient timestamp data. Studies were commonly comparative analyses, evaluating the performance of simpler statistical techniques with that of ML, or between linear and nonlinear ML modeling techniques. When simpler statistical techniques were included, ML methods demonstrated superior performance. Nonlinear models demonstrated improved performance compared with linear models. Within ML models, the incorporation of queue-based feature categories appeared to provide better performance in predictive estimates. Of note, research on the practical implementation of wait time predictions in the ED setting is lacking.

4.1. Good models are contingent on good data

Many studies highlighted dataset size as a limitation. When incorporating time-dependent features, as most studies did, the ability to capture long term dependencies will be contingent on having sufficient data from all time scales [36]. This is most relevant for studies with datasets spanning less than a year, as they might be excluding potential seasonal variability information. Another important consideration is that the data used to train a model should be representative of present workflows within the ED context. Including outdated workflow practices within the dataset will lead to inaccurate representations of model performance.

Among data quality issues raised, the most consequential was that the EHR timestamp record may not represent the reality of a patient's actual experience. From the authors' anecdotal experiences and as mentioned in Ang 2016 [21], physicians may 'sign-up' for multiple patients at a time but only see them in series, thus underrepresenting when they would see subsequent patients. This would skew the waiting time data, resulting in more underpredictions by a model, leading to decreased patient satisfaction as their expectations wouldn't match reality. It is therefore necessary to evaluate the veracity of the collected data from the EHR, to ensure that any model is trained on data that is reflective of reality.

The approach of focusing on low acuity or acuity-specific subsets of patients for wait time prediction in seven of the studies made logical sense given the perceived utility of the application. As pointed out in multiple studies [21,24,25,31], higher acuity patients will usually receive care quite quickly, leaving a wait time estimate not as useful for this subgroup. Furthermore, depending on the hospital, high acuity patients will sometimes be shunted down a different care pathway, limiting the acceptability of including them in the model training [25,31]. Lastly, Ang 2016 [21] mentioned that including data from these patients will lead to an underprediction for lower acuity patients. Hospitals looking to develop their own tool should therefore evaluate the wait times and care pathways of different severity patient populations to see if excluding high acuity patients makes sense for their implementation.

4.2. Queue-based features add the most predictive value

Across the studies that iteratively added additional feature categories, it was common to see the best model performance with the inclusion of all features. This suggests that waiting time was at least in part due to the influence of these multiple feature categories. However, because features were sequentially added as groups, individual-level feature influence wasn't discernable. This may have important implications in optimal feature selection, considering that some of these individual features could have been irrelevant or redundant. The inclusion of superfluous or redundant features may impact the speed of model learning, cause the model to overfit the training data, and increase computational costs [37]. Moreover, as Ang 2016 [21] mentioned, the complexity of the model and difficulty with which the features are retrieved from the EHR system increases as more features are added. This could be why time-based features were used the most between studies, followed by patient-based and queue-based features, due to their ease of retrieval. This is an important consideration when deciding what performance threshold would be practically meaningful. A balance might have to be struck between the desired performance of the model and the associated computational requirements.

Furthermore, the reviewed studies consistently identified queue-based features as significant predictors of wait times, or, as a set of features that can enrich the pool of predictors to improve the accuracy of predictions. For example, Ang 2016 [21] found between 8 and 35 % reductions in the test set MSE across the four different hospital datasets when queue-based variables were included in addition to time-based features. Benevento 2019 [13] also achieved test set MSE reductions of 23 % and 24 % between modeling techniques when queue-based predictors were included in addition to patient-based, time-based and staff/resource-based predictors. This was further confirmed in the Benevento 2023 [29] study, where prediction MSE was improved 8–18 % between datasets and models when queue-based and arrivals-based features were included. Arrivals-based features were estimates of the average number of arrivals per patient acuity and were grouped in this review under the queue-based features category. Lastly, Pak 2021 [25] determined a relative change of only 1 % mean squared percentage error when comparing queue only variables to their full feature model. These insights highlight an opportunity for further improvements in predictive wait time capabilities by incorporating patient queue information, assuming a hospital's data infrastructure can support real-time acquisition of the ED queue state.

4.3. The right model for the constraints of the context

The MAE, MSE, and RMSE evaluation metrics used aligns with other literature demonstrating these metrics as common primary metrics for machine learning regression evaluation [38]. Non-linear or 'black box' modeling techniques such as the Random Forest method, Gradient Boosting methods, LSTM, and Artificial Neural Networks demonstrated superior performance when compared against linear techniques. Similar results were found in [19], which demonstrated XGBoost to provide the best performance in predictive tasks. This supports the presence of nonlinearity in the interaction of predictive elements for a wait time in the ED. However, because of the greater computational costs in some cases and more restricted interpretability offered by these methods, it should be investigated whether the performance improvements justify these trade-offs. The Cheng 2020 [23] study for instance found practically negligible differences in MAE between the LSTM and linear regression model, but the LSTM had an over 40× longer training-testing time. In Benevento 2023, the 28 % and 17 % MSE reductions between the OLS and EM came at the expense of 660× and 570× increases in compute time. Moreover, comparing the MSE from the test sets of the Kuo 2020 [24] study between the best Linear Regression model and the non-linear models, a reduction of between 6 and 9 % was noticed, translating into a potential practical difference of only 3 min. Depending on the latency requirements and temporality of re-training for the implementation, linear techniques may be preferred. Furthermore, it is suggested that to increase healthcare professional and stakeholder trust and confidence in an AI-based decision, the approaches should be transparent, understandable and explicable [39]. This becomes more relevant when higher-stakes decisions are being made, and interpretability can enable proactive wait time management from staff if they can pinpoint the largest contributors to those values in real-time.

4.4. Model performance tailored to the end-user

In the studies reviewed, there were two distinct groups of end users that wait time predictions were intended for: patients and ED managers. Depending on the end user and the output modality, certain performance criteria should be considered during the model selection, training and evaluation process. Specifically, it should be determined whether an overprediction versus an underprediction of wait times may be preferred. There were no perfect averaged prediction estimates in any of the studies, and different modeling techniques would over- or underestimate waiting times to varying degrees. For instance, [21,25,27] emphasized the propensity of rolling average methods to underpredict when waiting times were long and overpredict when waiting times were short. Pak 2021 [25] also determined quantile regression to be an effective method for mitigating underpredictions. Moreover, [27,29] commented on the Random Forest method being more robust to underpredicting.

From the patient perspective, wait time predictions could be displayed on a hospital's website or in the physical ED. For the former, an overprediction of wait times may divert patients who would have otherwise come to the ED, causing them to miss out on potential required treatment. Queuing theory describes this as a concept called balking, where a customer who has not yet joined the queue decides not to join due to the queue length [40]. This would come with additional financial ramifications to the ED, as revenue associated with that patient is lost. Because of this, a modeling strategy less susceptible to overpredictions, such as Lasso or Q-Lasso, may be preferred [21,25]. In the latter example, balking is less of a concern since the patient has already arrived. An overprediction may be ideal in this case, supported by the expectancy-disconfirmation effect, which states that a person experiences higher satisfaction when expectations are exceeded [41–43]. This becomes realized if patients are seen earlier than they had expected from the prediction. The study by Ang 2016 [21] echoed this sentiment, where an early-stage prediction implementation alluded to underpredictions leading to reduced patient satisfaction. Therefore, a Random Forest or Quantile Regression method may provide more benefit to patient satisfaction in this modality. A balance between overpredicting and accuracy may be most appropriate, however, as highly inflated wait times could lead to more patients leaving without being seen [11]. An in-depth understanding of patient waiting tolerance is thus necessary to effectively adjust the threshold of acceptable overprediction.

A tradeoff also exists between under- and overprediction from the ED management perspective. An overprediction of wait times could lead to an excess of staff resources when these estimates are acted upon, benefiting patients but incurring unnecessary costs to the hospital. This would result in more on-call staff being recruited to alleviate a perceived system overload. Such situations should be minimized considering on-call staff shifts are associated with negative health outcomes and family complications [44,45]. Conversely, the underprediction of wait times would minimize staff resources and cost but prevent the decongestion of an overcrowded ED. To work well in practice, decision makers need to understand the modeling limitations and error rates, providing them with the knowledge to make informed decisions. As described in [46], understanding can be enhanced when model interpretability is conducted in a human-friendly way. Conveyance of model performance should therefore emphasize MAE and MAPE metrics as these are the most intuitive for a non-technical audience.

Lastly, most of the studies presented the wait time predictions as point estimates. Arora 2023 [30], however, had argued that point estimates can be uninformative and misleading because of the inherent uncertainty and asymmetric distribution of waiting times. They instead used a Quantile Regression Forest to present probabilistic estimates. Conveying the predictive uncertainty in the form of detailed probabilistic estimates could facilitate systematic stakeholder trust, improving the acceptability of the implementation [47]. More research is needed on this topic, and it may be beneficial to include end users during the development process to identify their preferences.

4.5. Ethical and regulatory considerations

Applying wait-time predictions in an emergency setting that may influence patient and staff behavior requires cautious development transparency and implementation to avoid undue harm to individuals. The European Union AI Act, proposed by the European Parliament and Council of the European Union [48], was the first of its kind to provide a regulatory framework behind societal implementations of AI, and provision development standards for high-risk AI systems such as ED wait time predictions. [49] outlines limitations in current cybersecurity prevention for AI systems that must be considered when implementing high-risk systems to satisfy the EU AI act conditions. This scoping review identified shortcomings in the bias and transparency of model development within some articles. For instance, just over half of the papers provided detailed data preprocessing steps for dropped column proportions, and some didn't include out-of-sample test set results to establish real-world performance. Furthermore, including performance results stratified by patient cohorts could have bolstered the trust in model capabilities for all patients, like in [35]. Addressing these concerns will aid in the regulatory compliance of ML in healthcare, while fostering increased trust and uptake of these tools in the clinical setting.

4.6. Future directions

These studies have established AI as a viable option for improving patient wait time predictions in an ED, but the extent to which the return on investment for the time dedicated to implementing these solutions should be explored. Related questions could include what minimum performance in accuracy is needed to derive practical benefit to the hospital, and if there are additional advantages to further improving the prediction accuracy. This could be adapted through the lens of either the patient's experience or hospital management decision-making. Moreover, effective strategies for determining when to act upon a wait time prediction estimate from the hospital management perspective are underexplored.

Furthermore, the reviewed studies lacked long-term implementation feasibility assessments. Future work should evaluate what minimum data infrastructure is required to perform such functionalities, on what timescale model re-training should be conducted, and best practices for facilitating the implementation success and sustainability of these tools. Tsoni et al., [50] suggest using open-source user friendly tools, such as KNIME, to streamline the data preprocessing and ML pipeline for low-code users. Additionally, it should be deduced what level of uncertainty presentation, if any, is preferred from the user perspective to enhance the predictions acceptability and understanding. Lastly, having accurate wait time estimates could enable low acuity patients to 'sign up' for a queue from the comfort of their home, arriving at the hospital when the doctor is predicted to be ready for them. This would alleviate congestion within the ED as patient flow would become more consistent, although the feasibility of this concept should be investigated further.

4.7. Limitations of the studies included in the review

Minor methodological inconsistencies were noticed throughout some of the papers reviewed. Primarily, in certain instances, an explicit statement defining how the waiting time was derived was missing or unclear. Furthermore, evidence-based decisions for including features in a model were sometimes not available. Lastly, the results tables within some studies caused confusion as to whether training, validation or testing set performance was being referenced. By mitigating these minor inconsistencies, the understanding and transferability of research on this application of AI could be enhanced, enabling others to more easily replicate findings.

4.8. Limitations of the scoping review

Limitations of the scoping review include variations in terminology, language barrier exclusions, and the complexity of studies. Specifically, varied terminology for similar concepts may have led to incomplete search results and/or results may not account for all possible nomenclature variations. Moreover, studies that were not conducted in English were excluded, which could have limited the scope of the search and potentially missed relevant study data. Finally, the complexity in methodological conduct of a study may have led to difficulties interpretating and aggregating the data for the authors who didn't have as much domain expertise.

5. Conclusion

We identified and explored some of the methodologies conducted thus far using AI to predict patient waiting times in an ED. Most were comparative analyses of linear and nonlinear modeling techniques, using detailed retrospective EHR data from one or multiple hospital sites. The most prevalent feature category affecting performance appeared to be queue-based features, emphasizing the importance of capturing ED queue-state dependencies to improve prediction accuracy. Random Forests and Linear Regression modeling were the most used, with nonlinear techniques demonstrating superior performance across the studies identified. Gradient Boosting Machines were found to attain the best performance in each study they were used. However, the performance benefits of more complicated machine learning models come at the expense of greater data and computation requirements. These are considerations for practitioners and hospital management to keep in mind as they decide which strategy to pursue.

Additionally, part of the nuance involved in wait time prediction surrounds where, how, and to whom the prediction is being applied. Different modeling techniques will over- and under- predict to varying degrees with either direction being preferred depending on the context and the end-user. It is up to relevant stakeholders to decide which is preferred, depending on their specific goals and requirements. Furthermore, more work is necessary to delineate what performance accuracy threshold of predictions will lead to meaningful outcomes. Regardless, this review consolidated the supportive evidence that the use of AI in predicting patient wait times can exceed the performance of traditional simpler statistical methods such as rolling averages. Providing a succinct resource for practitioners and hospital management to reference will hopefully streamline efforts to implement site-specific solutions and ultimately reduce the global burden of ED overcrowding.

CRediT authorship contribution statement

Troy Gloyn: Writing – original draft, Visualization, Validation, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Christina Seo:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Alexandra Godinho:** Writing – review & editing, Validation, Project administration, Methodology, Data curation, Conceptualization. **Rahul Rahul:** Writing – review & editing, Validation, Data curation. **Siona Phadke:** Writing – review & editing, Validation, Data curation. **Hilary Fotheringham:** Writing – review & editing, Validation, Data curation. **Pete Wegier:** Writing – review & editing, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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